

Use Cases

Name: Using an elevator

Primary Actor(s): Passenger(s) of the elevator

Stakeholder and Interests:

Customer(s)/Passenger(s) - is using the elevator directly for transportation

Owners of building the elevator is in - wants elevator to be safe and is responsible for any unsafe scenarios that occur

Safety regulators/inspectors - responsible for the regulations the elevator(s) is built with and performing the final inspection that allows the elevator(s) to be used

Precondition(s): passenger(s) are inside the building the elevator is in

Success guarantee(s) (equivalently Post-conditions): passenger(s) exited the elevator

Main success scenario:

1. passenger has pressed the up or down button
2. button is illuminated until elevator arrives, going in direction that the customer has requested
3. a bell rings and the door is opened for 10 seconds, the passenger(s) enter and exit the elevator during the duration the door is opened, until the bell rings and the door is closed
4. the on-board passengers select their destination floor using the panel
5. the elevator travels to each floor requested, while displaying current floor it's on in a display and using a sensor to notify when it arrives at a floor

Extensions:

- 3a. the "open door" button on the elevator control panel is depressed
 - 3a1. the elevator doesn't move and the doors remain open (beyond the default timing) until the button is not depressed
- 3b. the "close door" button on the elevator control panel is pressed
 - 3b1. the doors close prematurely
- 3c. the light sensor is interrupted when the door is closing
 - 3c1. refer to use case *The light sensor is interrupted*
- 3d. the elevator is overloaded
 - 3d1. refer to use case *The "overload" alarm*
- 4/5a. "help" button is pressed
 - 4/5a1. refer to use case *The "help" alarm*
- 4/5b. there's a fire
 - 4/5b1. refer to use case *The "fire" alarm*
- 4/5c. the power goes out
 - 4/5c1. refer to use case *The "power out" alarm*

Name: The "help" alarm

Primary Actor(s):

Passenger(s) of the elevator

Building safety service

Emergency services

Stakeholder and Interests:

Customer(s)/Passenger(s) - wants to be helped

Owners of building the elevator is in - responsible for building safety services

Safety regulators/inspectors - responsible for ensuring the elevators in the building have the appropriate protocols for the help signal

Building safety service - 1st responders responsible for helping the passengers

Emergency services - responsible for helping the passengers if building safety services can't

Precondition(s): there are passengers inside the elevator

Success guarantee(s) (equivalently Post-conditions): the passengers have successfully connected to either building safety services or a 911 emergency call

Main success scenario:

1. the "help" button is pressed by a passenger
2. the passenger is connected to building safety service through a voice connection

Extensions:

- 2a. building safety service does not respond within 5 seconds
 - 2a1. emergency services (911) is automatically called by the elevator

Name: The light sensor is interrupted

Primary Actor(s): Passenger(s) of the elevator

Stakeholder and Interests:

Customer(s)/Passenger(s) - is directly at risk by using the doors

Owners of building the elevator is in - wants no accidents with the door to occur in their property

Safety regulators/inspectors - responsible checking that the elevators follow elevator door regulations

Precondition(s): the door is open

Success guarantee(s) (equivalently Post-conditions): the door is closed

Main success scenario:

1. the door is closing
2. some sort of obstacle between the door interrupts the light sensor
3. the control system stops the door and opens it
4. the door closes

Extensions:

- 3a. the door is interrupted repeatedly over a short period of time
 - 3a1. sound a warning over the audio system
 - 3a2. display a text message warning

Name: The "fire" alarm

Primary Actor(s): Passenger(s) of the elevator

Stakeholder and Interests:

Customer(s)/Passenger(s) - wants to be outside the elevator in a fire

Owners of building the elevator is in - responsible for elevators to follow fire protocol

Safety regulators/inspectors - responsible for ensuring that the elevators follow fire protocol during inspection

Fire department - wants passengers to be in positions that minimize risk (i.e. not inside an elevator) when there is a fire

Precondition(s): the building fire alarm signal is sent to the control system or signaled by the elevator itself

Success guarantee(s) (equivalently Post-conditions): the passengers are on a safe floor, outside the elevator

Main success scenario:

1. the elevator goes to a safe floor
2. an audio and text message informs the passengers of an emergency and to disembark

Extensions:

Name: The "overload" alarm

Primary Actor(s): Passenger(s) of the elevator

Stakeholder and Interests:

Customer(s)/Passenger(s) - wants the elevator to function normally

Owners of building the elevator is in - responsible for elevators to adhere to specific weight limits

Safety regulators/inspectors - responsible for ensuring that the elevators follow regulations with weight limits

Precondition(s): the elevator is open

Success guarantee(s) (equivalently Post-conditions): the elevator resumes normal function

Main success scenario:

1. sensors detect an "overload" signal after the passengers step in and the total weight exceeds the carrying capacity
2. elevator is stopped with the door open
3. audio and text messages presented to passengers asking for reduction in load
4. elevator attempts to move by checking the capacity again

Extensions:

- 4a. the capacity is still over the limit
 - 4a1. repeat step 3 (ask the passengers to reduce the load)
 - 4a2. attempt the move the elevator again

Name: The "power out" alarm

Primary Actor(s): Passenger(s) of the elevator

Stakeholder and Interests:

Customer(s)/Passenger(s) - wants to exit an elevator that has no power

Owners of building the elevator is in - responsible for backup power and the protocol that elevators without power should follow

Safety regulators/inspectors - responsible for ensuring that the elevators follow regulations with "power out"

Precondition(s): the elevator is not able to receive normal power in some way

Success guarantee(s) (equivalently Post-conditions): the passengers are on a safe floor

Main success scenario:

1. the control system receives a "power out" alarm signal
2. the elevator uses backup power
3. audio and text messages inform passengers of the power outage
4. the elevator moves to a safe floor
5. the passengers are asked to disembark via audio and text message

Extensions: